Analysis & Implementation of Ultra Low-Power 4-bit CLA in Subthreshold Regime

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***Abstract*--The paper presents the analysis and implementation of ultra low-power, low voltage and low area 4-bit carry look ahead adder circuits. Sub-threshold design technique has been used to reduce the power consumption and area while maintaining low complexity of logic design in the proposed circuit. Simulation results illustrate the superiority of the circuits in sub-threshold region against the conventional low power design technique, in terms of power, area and power delay product (PDP). The CLA is implemented on TSMC 0.18µm process models in Cadence Virtuoso Schematic composer with improved driving ability and circuit robustness at 0.4V single ended supply voltage and simulations are carried out on Spectre S. The proposed 4-bit CLA can operate up to 5 MHz and used 0.035 µW of power and occupied an area of 60x92.5 µm^2.**

Keywords—Carry look ahead adder (CLA); Power delay product (PDP) ; sub-threshold; Carry ripple adder(CRA)

I. INTRODUCTION

With the rise of portable, battery operated devices reducing power consumption has become an important design goal. Increased power consumption also leads to increased heat, which makes low power design important even for non-portable applications. Circuit designers now seek to meet performance requirements for a device with the lowest power consumption possible. Digital circuits operating in the sub-threshold region of the transistor are being used as an ideal option for ultra low power CMOS design. To operate a circuit in sub-threshold regime the operating voltage scaled down below the device threshold [1].

Energy efficient adders under sub-threshold logic are essential part of microprocessors and systems on chip (SOCs) modules. Thus, optimizing the adder circuitry to reduced power consumption will have a beneficial effect on many types of digital modules. Carry- Look ahead Adder (CLA) is fast adder architecture optimized at the logic-level, and used to design a high performance addition-based arithmetic circuits using cell based VLSI design [2]. Many design architecture and techniques have been developed to reduce power dissipation complementary logic. In [3], a 4-bit energy efficient CLA has been implemented using dynamic logic and optimized Power delay product in femtoJoules (fJ) range. In [4], a new family of self-resetting logic primitive gates and adder cells is implemented which has good PDP as compared to conventional dynamic logic circuits. In [5], low power 4-bit Carry-Ripple and Modified Carry Look-Ahead adders cells based on sub-threshold logic in nanometer CMOS technology is presented. The simulation result shows that the MCLA has a lower delay, power-delay and energy-delay products as compared to CRA. In [6], the low-power carry look-ahead adder using MTCMOS was designed. The propagation delay time is reduced by using low threshold voltage transistor in the critical path.

In this paper, energy efficient CLA adder components are designed and analyzed using static CMOS in sub-threshold region. The functional requirement of the adder is, to be able to perform unsigned addition of two 4-bit numbers in parallel. Power consumption must also be minimal. A bottom-up design methodology is adopted in the development of the 4-bit CLA. The fundamental components are first designed and optimized such as Propagate/ Generate generator, sum generator, and carry generator. Secondly, the components are integrated together into a final schematic and optimized using various techniques that is mentioned in their respective sections. Thirdly, the design is translated into a layout and simulation is once again conducted. Lastly, comparison of pre layout simulation waveform and analysis result is being discussed.

II. LOGIC AND CIRCUIT DESIGN OF SUBTHRESHOLD 4-BIT CLA ADDER

In this section we consider the basic logic and circuit design of sub-threshold 4-bit CLA. The propagation delay is also be optimized in CLA by calculating the carry signals in advance, based on the input values. This adder is based on the principle of looking at the lower order bits of the augend and addend if a higher order carry is generated. This adder reduces the carry delay by reducing the number of gates through which a carry signal must propagate. This adder consists of three stages: a propagate block/ generate block, a sum generator and carry generator. The generate / propagate block, sum generator and carry generator can be realized using the expression (1)- (8) [7].

$$Gi = Ai.\ Bi \quad \text{---------- (1) Carry generate.}$$

$$Pi = Ai \oplus Bi \quad \text{-------- (2) Carry propagate.}$$

Notice that both propagates and generates signals depend only on the input bits and thus will be valid after one gate delay. The new expressions for the output sum and the carryout are given by:

$$Si = Pi \oplus Ci \quad \text{---------- (3)}$$

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$$C_{i+1} = G_i + P_i.C_i \quad \text{---------- (4)}$$

The final equations of 4-bit CLA adder is defined as:

$$C1 = G0 + P0C0 \quad \text{---------- (5)}$$

$$C2 = G1 + P1C1 = G1 + P1\,(G0 + P0C0) = G1 + P1G0 + P1P0C0 \quad \text{---------- (6)}$$

$$C3 = G2 + P2C2 = G2 + P2G1 + P2P1G0 + P2P1P0C0 \quad \text{-- (7)}$$

$$C4 = G3 + P3C3 = G3 + P3G2 + P3P2G1 + P3P2P1G0 + P3P2P1P0C0 \quad \text{---------- (8)}$$

The proposed CLA circuit comprises of XOR, NAND and Inverter logic gates which employed the sub-threshold static CMOS logics. The 4-bit CLA consist of propagate/ generate block, carry generator block, sum blocks as shown in Fig1. The basic block diagram of CLA circuit has inputs Ai and Bi which goes into propagation / generation blocks and the output of this block goes as the input of the carry generation unit or look-ahead block. The carry bits are XORed with respective propagation bits to give the final output bits of the CLA. In fig 2 the transistor level circuit diagrams of carry generator block i.e. C1, C2, C3, C4 is shown.

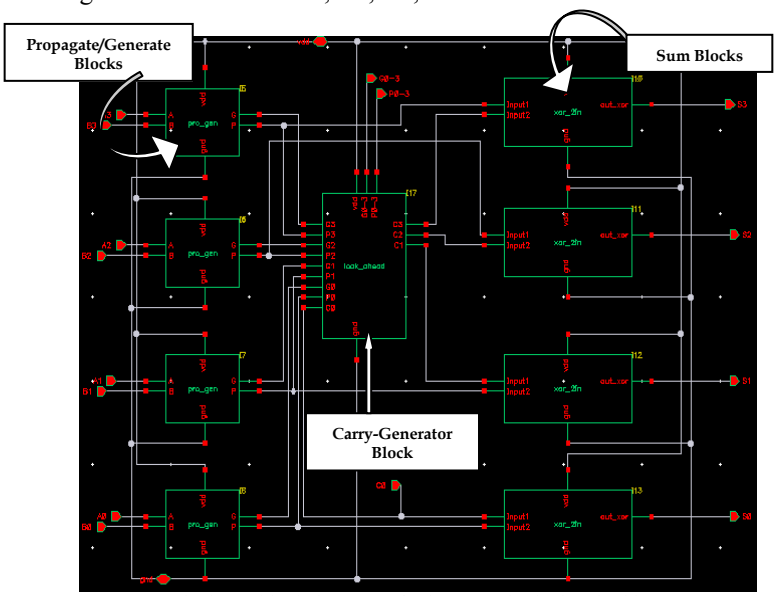


Fig1. The basic block diagram of 4-bit carry look ahead adder circuit

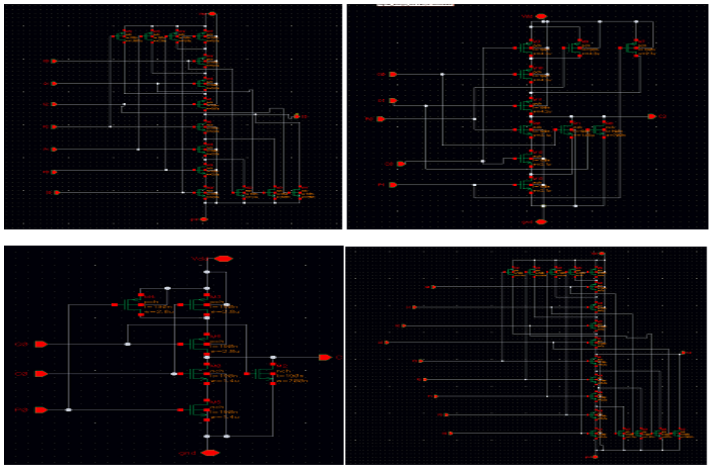


Fig2: Circuit diagram of 4-bit carry generator C1,C2,C3,C4

III. LAYOUT DESIGN OF THE SUBTHRESHOLD 4-BIT CLA

The 4-bit CLA schematic designed into pre-fabrication layout. The design methodology employed is once again bottom-up. Since whole schematic are the composition of basic three modules i.e. propagate/ generate block, carry generator block, sum block , so the layout of these basic blocks are completed and tested first before integration into the final 4-bit CLA. The complete layout design and their carry generator circuit layout of 4-bit CLA is shown in fig3 and fig4. The implemented 4-bit CLA has occupied an area of 60x92.5 μm^2.

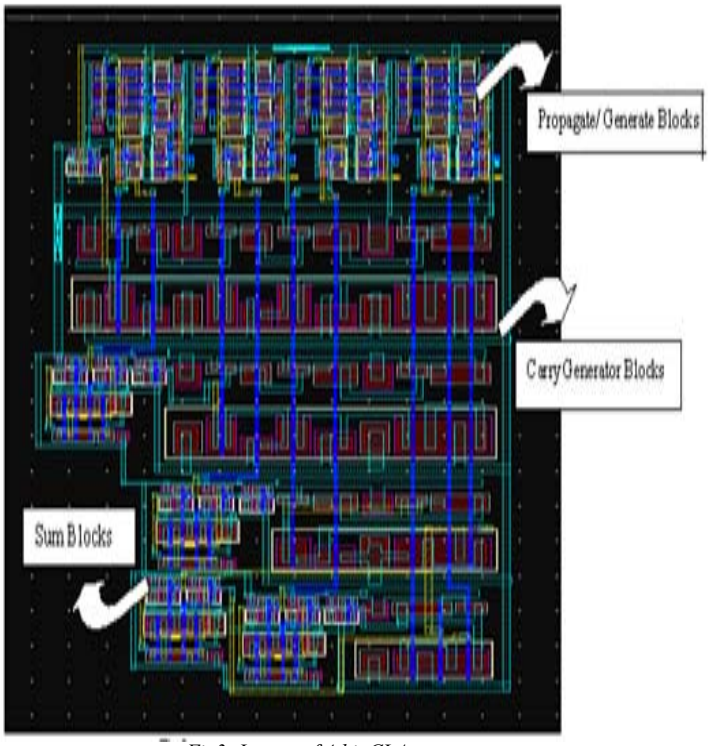


Fig3: Layout of 4-bit CLA

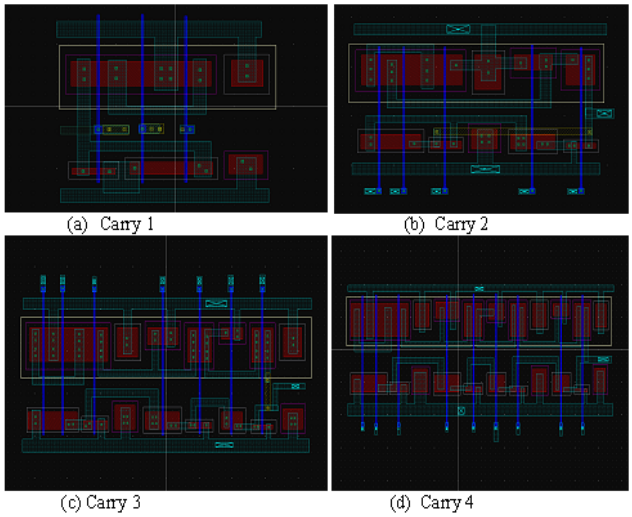


Fig. 4 Layout of carry generator (a) Carry1 (b) Carry2 (c) Carry3 (d) Carry4

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IV. PERFORMANCE ANALYSIS OF THE 4-BIT CLA

A parametric analysis on varying the PMOS transistor width was done to observe the power consumption. The power consumption is least when we have minimum size for both NMOS and PMOS. As per the conceptual analysis to allow the same current in both PMOS and NMOS, the W/L ratios of PMOS and NMOS should be in the inverse ratio of the mobility ratios of hole to electron. So to have a symmetrical design, the W/L ratios of PMOS and NMOS are taken in the ratio 2. The 4-bit CLA has been simulated in the 180nm technology at various frequencies ranging from 1 KHz to 5 MHz with the supply voltage of 0.4 V. For having more realistic simulation condition, buffers are used before all the inputs. The simulation results for the 4-bit CLA at frequency of 5 MHz are shown in Table I. The results are compared with the existing low power CLA in terms of the power consumption, delay, power delay product and area.

4-bit CLA	Power Dissipation (µW)	Delay (ns)	PDP (fJ)
Ref [1]	210.01	0.06	12.59
Ref [2]	156780.00	0.136	21320.00
Ref[3]	0.117	48.66	5.6961
Ref[4]	1160	2.39	2770.00
Proposed CLA	0.035	50	1.75

Table I: comparison table

The result shows that proposed CLA circuit provides an improved overall performance in terms of speed, area, and power dissipation. Table I. shows the comparison of existing CLA with proposed one in terms of PDP, propagation delay, power dissipation and area.

V. SIMULATION RESULT OF SUBTHRESHOLD 4-BIT CLA

The proposed 4-bit carry look-ahead adder is simulated through *Spectre* simulation using the 180nm CMOS technology with a supply voltage of 0.4V. The average power dissipation for the circuit is measured as 0.035 µW at 5 MHz clock frequency. A series of test vectors is used to test the adder operation. The tests proved the adder to be fully functional. With the functionality verified another test was performed to determine the maximum operating speed. The CLA is operated in 5 MHz range. In fact, in addition to normal transistors, implemented circuits are tested in corner cases with fast and slow transistors and their combinations too. Fig. 5 shows the output waveform of 4-bit CLA.

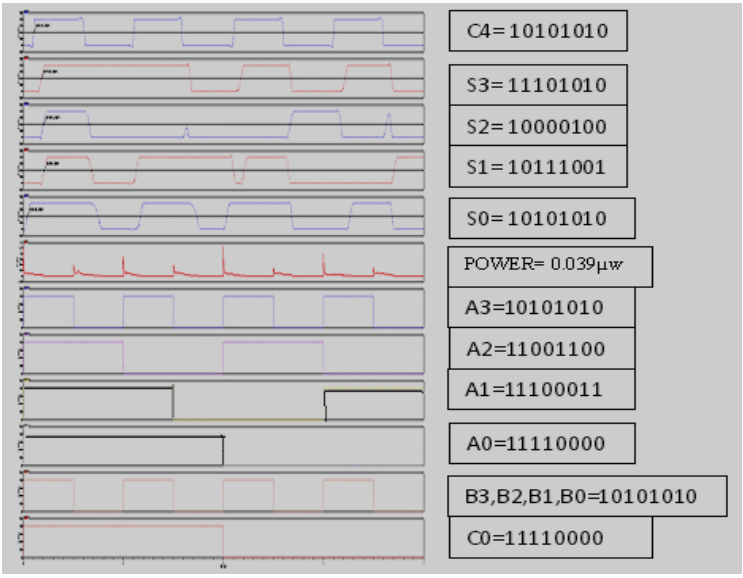


Fig. 5 Output Waveform of CLA

VI. CONCLUSION

In this work, low power 4-bit Carry Look-Ahead adders based on sub-threshold logic styles were presented. The 4-bit CLA were constructed based on static CMOS logic style. To compare the performances of the proposed adders, different types of logic families based CLA has been explored. The study includes the effect of changing the supply voltage, sizing of the transistors and frequency. The results show that proposed CLA circuit provides an improved overall performance in terms of speed, area, and power dissipation. The designed 4-bit CLA adder operated at 5 MHz and used 0.035 µW of Power and occupied an area of 60x92.5 μm^2.

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